

**Instructions (read these carefully!):**

- This assignment must be submitted on Gradescope by the due date. Gradescope entry code: PRXD46
- Please **include a title page** with your name and the names of your collaborators.
- Homework is graded anonymously, so please avoid putting your name on each page of the assignment.
- If you are stuck, please ask for help (from me, Sarah, a classmate, ...). Use Ed discussions! AI use is generally **not** recommended for completing the homework.
- You may freely use any fact proved in class (mention it in the appropriate spot). You *may not* freely use facts from the book or other sources.
- As a general rule, try to restrict your solution to each problem to a single page. Often solutions can be short, while still being complete. If your solution is very long, you should think more about how you might express it more concisely.

**HW4: harmonic conjugates, complex integration**

Due: Friday, Oct 9 by 5pm

1. True or false: the set  $\mathbb{C} \setminus \mathbb{R}$  is simply connected. (Warning: the definition of simply connected that I gave in class was very likely<sup>7</sup> incomplete.)
2. Compute from the definition.<sup>8</sup>
  - (a) the (MVC) line integral  $\int_{\gamma} f$  where  $f(x, y) = (x, y)$  and  $\gamma(t) = (t, t)$  for  $t \in [0, 1]$ .
  - (b) the complex line integral  $\int_{\gamma} f$  where  $f(z) = z$  and  $\gamma(t) = t + it$  for  $t \in [0, 1]$ .
3. Show that if  $D$  is a bounded region with differentiable boundary, then

$$\int_{\partial D} \bar{z} dz = 2i \text{ area}(D).$$

4. Show that if  $u : U \rightarrow \mathbb{R}$  is harmonic and  $\gamma : [0, 1] \rightarrow U$  is a simple, closed curve that bounds a disk in  $U$ , then  $\int_{\gamma} u_x dy - u_y dx = 0$ .
5. Use the preceding problem to show that a harmonic  $u$  on a simply connected  $U$  has a harmonic conjugate on  $U$ .<sup>9,10</sup>

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<sup>7</sup>There are two conditions (corresponding to the two words in the name). I often key in on the more interesting condition, forgetting to mention the other one.

<sup>8</sup>Note for these two examples, you are integrating the same function, over the same path!

<sup>9</sup>Hint: fix basepoint  $(x_0, y_0)$  and define  $v(x, y) = \int_{\gamma_{x,y}} u_x dy - u_y dx$ , where  $\gamma_{x,y}$  is a path from  $(x_0, y_0)$  to  $(x, y)$ . You should check that this is well defined.

<sup>10</sup>Hint: Check CR equations. For example compute  $v_x$  using the definition of the partial derivative and well-chosen paths for computing  $v(x + h, y)$  and  $v(x, y)$ .